1 City graph

cells with price group, existing-line flag and OD degree; spatial edges (8 neighbours) and OD-flow edges

2 Graph encoder

two message-passing layers produce an embedding h for every cell; computed once per update for all 16 episodes

3 Weight vectors

4 vectors w per update from Dirichlet(1,1,1); 4 rollouts for each. w tells the policy which trade-off to build for

4a Start head

scores every cell as the first station; query uses the mean embedding and w; plus the fixed scalarised prior

4b Move policy

scores the 8 neighbours. key: embedding h_c and look-ahead features (one-step gain per group, new reach, line adjacency, stations left); query: line pool, w, group shares

5 Rollout

start, then up to 19 moves under the build rules (inside grid, no revisit, near-straight line)

6 Objectives of the line

demand D, fairness G and accessibility A, each divided by the best single-objective value found by a genetic search

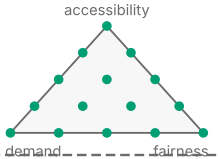
7 Return and advantage

R = -g(f | w), augmented Tchebycheff. Advantage = R minus the mean R of the other three rollouts with the same w

8 Update

advantage-weighted log-likelihood of every decision, entropy bonus, one Adam step; no replay buffer, no critic

new parameters



Evaluation (after training, same for every method)

15 weight vectors on a simplex lattice. For each one the method hands over one line (greedy rollout, or best member of the final population): the deployable set. All lines evaluated during the run form the archive. Both sets are scored by hypervolume and expected utility over the 15 weights, on the scaled objectives.